

Research Proposal

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1. Background

Government often acts as the largest buyer of the economy through government procurement. In China, the scale of government procurement is steadily expanding, hitting 3.7 trillion yuan (about 572.5 billion US dollars) in 2020. Understanding how government procurement impacts broader economic outcomes through firm performances, like innovation, is essential for policy design. This proposal focuses on the following research question: How does obtaining government procurement contracts affect firms' subsequent innovation?

This research exploits a novel contract dataset to identify the causal effects of winning government procurement contracts on firms' subsequent innovation. Specifically, I'll use a quasi-exogenous context of "close auctions" (i.e., winners and runners-up have very close scores and the contract allocation to first two bidders is therefore quasi-random), to examine the differential dynamics of winners and runners-up after the auction. I'll first identify the "close auctions" defined with a small win margin; then, I'll restrict the sample to winners and runners-up who have not previously won any government contract; next, I'll use an event study design to examine the dynamic effects of obtaining the first contract on firms' subsequent innovation, mainly measured with patent applications. I'll further conduct rich mechanism analysis to investigate several possible channels, say signaling effect, capital provision, R&D investment, and political connections.

This research will mainly contribute to two strands of literature. The first literature is about how government procurement impacts firms' economic outcomes. Both existing empirical evidence and theoretical frameworks lead to ambiguous effects. On the one hand, government contracts could have positive signals for firm quality or future cash flow, thus attracting external funding, high-skill workers, and customers (Howell, 2017; Lerner, 1999; Goldman, 2020; Gabriel, 2024); or provide government resources that are shareable for commercial uses, bringing about economies of scale (Beraja et al., 2023). On the other hand, however, government contracts always emphasize lower prices, squeezing the profit space of firms (Cao et al., 2022); government contracts also prioritize off-the-shelf goods, with limited requirement of innovation (Howell et al., 2021); in addition, government is found to selectively delay payment to small and medium enterprises during China's deleveraging, worsening their financial conditions and operating performance (Hu et al., 2025). These factors are in turn discouraging innovation. This research uses a quasi-natural experiment of contract awarding in close auctions to better understand the causal effects and mechanisms on the topic.

This research is also related to the literature of innovation policies in China. Innovation policies are typically grouped into input-side policies that lower the cost of innovating and output-side policies that raise the returns to innovation. Input-side policies include subsidies (Li, 2012; Kou & Liu, 2020), tax cuts (Chen et al., 2021; Shao & Xiao, 2019), etc. Output-side policies, such as stronger IP protection, also stimulate innovation in China (Fang et al., 2017; Wu & Tang, 2016). This research focuses on the indirect consequences government could exert on innovation, through government procurement contracts, by affecting both the input side (e.g., government resources) and the output side (e.g., guaranteed market demand).

2. Data

This research mainly uses three datasets. **(i) Procurement Contract Dataset:** The data is scraped from the government procurement website of one Chinese province. It includes detailed records of all government procurement contracts in the province, containing rich information on the procuring entity, participating firms, bid scores (technical and price components) for all bidders, and bid value. In total, the data cover over 300,000 procurement projects from 2015 through 2025, with the bid scores available from 2021 onwards. **(ii) Firm-level registration data:** In order to include a rich set of controls, as well as to do mechanism analysis, I utilize the State Administration of Industry and Commerce (SAIC) database in China. The SAIC provides complete records of firm names and domiciles, registered capital, business scope, shareholders, year of establishment, etc. **(iii) Firm-level patent Data:** To capture firms' innovation outcomes, I assemble data on patent applications and grants, as well as R&D inputs where available, sourced from the database of the China National Intellectual Property Administration (CNIPA), formerly SIPO.

3. Empirical Design

3.1 Baseline Specification

This research investigates the impact of winning procurement contracts on firms' subsequent innovation, using a rich contract dataset containing evaluation scores for all bidders. I'll identify the causal effects in a quasi-exogenous context of "close auctions", i.e. winners and runners-up have very close scores and the contract award is therefore quasi-random¹. Specifically, I'll identify close auctions in all procurement contracts defined by the win margin², and retain only winners and runners-up who haven't previously won an award. The use of close auctions is critical for the empirical design, largely mitigating selection concerns. Imagine one directly compare winners and losers, there could be severe bias, like politically connected firms may be both more likely to win an award and poorer in innovation. I also focus on firms without previous awards to get a more homogeneous sample, as well as to isolate the effect of the first contract. The baseline specification is given as follows:

$$y_{it} = \sum_{T \neq -1} \beta_{1T} (T_{it} \times Winner_i) + \sum_{T \neq -1} \beta_{2T} T_{it} + \sum_{T \neq -1} T_{it} \times X_i + \gamma_i + \lambda_{ct} + \delta_{jt} + \epsilon_{ijt},$$

The outcome variable, y_{it} , is firm i 's cumulative innovation output up to period t , measured by the total number of patent applications (or $\log(1+\text{patents})$). The explanatory variables of interest are the interaction terms between a full set of time period dummies, T_{it} , and $Winner_i$. Specifically, T_{it} is a set of dummy variables indicating whether period t is T periods before or after firm i 's first auction, with $T = -1$ omitted as the reference period; $Winner_i$ equals one if firm i wins that auction and zero

¹ This is based on the evaluation process of government procurement through open bidding, where at least three firms submit bids and the contract is awarded to the bidder with the highest overall score (based on both product quality and price); the scores are evaluated independently by at least 3 anonymous experts, and are not disclosed before the award. Therefore, the contract award cannot be manipulated by any expert or firm, making it quasi-random between top two bidders. One concern is the possibility of score rules manipulation, but this form of manipulation would typically reflected in a large win margin, which I've excluded in the study sample.

² For example, when $\text{score_margin}_k = \frac{\text{winner_score}_k - \text{second_score}_k}{\text{second_score}_k} < 0.5\%$. The specific bandwidth is guided by the empirical distribution of margins and is allowed to vary in robustness checks.

if it is the runner-up. The coefficients on $\sum_{T \neq -1} (T_{it} \times \text{Winner}_i)$ trace the dynamic effect of obtaining a first government procurement contract on subsequent patenting, comparing close-auction winners to runners-up. I include a vector of pre-auction firm characteristics X_i (e.g., firm age, pre-event patent applications, etc). To absorb all time-invariant firm characteristics that may jointly affect bidding outcomes and innovation, I add firm fixed effects γ_i . I also include city-by-time fixed effects, λ_{ct} , to control for time-varying local shocks, such as city-specific procurement practice and innovation policy. Industry-time fixed effects, δ_{jt} , are added to absorb time-varying shocks to different industries. Standard errors are clustered at the auction level, allowing arbitrary within-auction correlation in the error term. As a robustness check, I'll also use two-way clustered standard errors by firm and auction to allow for both within-firm serial correlation and within-auction correlation.

The identifying assumption is that, the winners and runners-up would have followed parallel innovation trends before the auction. This is credible because, contract award in close auctions is quasi-random, so winners and runners-up are comparable; and I've controlled a set of pre-auction firm characteristics, further mitigating the concern that some dimensions may not be fully captured by bidding scores. I'll test the assumptions by checking whether estimated β_{1T} are statistically indistinguishable from zero, for all $T < 0$.

3.2 Mechanism Analysis

Signaling Effect Winning government contracts may send a positive signal for outside investors, thus attracting external funding, which are especially essential for the innovation of high-tech firms. To test this channel, I'll further restrict the sample to high-tech firms, and interact the $T_{it} \times \text{Winner}_i$ with pre-event measure of local VC intensity (e.g., VC deals per capita, share of VC-backed firms in the city), to investigate whether the relative effects are more positive for high-tech winners in VC-intensive regions.

Capital Provision Government contracts can also facilitate innovation through direct capital provision, i.e., cash flow, credit access. I'll interact the main treatment term with the contract value. If capital provision is important, the post-auction innovation effect should be more positive for firms that won higher-value contracts.

R&D investment To investigate whether winners have invested more in R&D or hire more research staff, I'll replace the outcome variable with cumulative R&D investment (share of total sales, for example) and additional research staff employment.

Political Connections It's also possible that early-stage winners take one step faster to build political connections with procurement agency, and secure more contracts in the future. I'll first check if winners will participate in more collusive bids in subsequent auctions, following the method in Chen (2025) to detect corrupt procurement contracts. Next, I'll check whether winners have higher probability of winning subsequent contracts, by replacing the outcome variable with the share of contract award in overall auction participation. If winners conduct more collusive bids, and become

more likely to secure subsequent contracts, it may indicate that winners are better politically-connected, which is expected to amplify the first-contract effect.

References

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